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Agentic AI for Autonomous Cybersecurity Threat Mitigation

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Abstract- The increasing sophistication of cyberattacks is requiring dynamic and autonomous defense processes that have the capacity to prevent, investigate, and suppress threats. This research will introduce an agentic Artificial Intelligence (AI) model of autonomous cybersecurity threat mitigation that relies on the Reinforcement Learning (RL) as the main algorithm and the use of the TensorFlow as tools. The framework relies on the RL agents that continuously train the optimal defense policies with the help of interaction with the dynamic cyber environment, as well as recognizing and applying real-time countermeasures on its own. The model also enhances the accuracy of detection, reducing false positive and maximising response time due to the scalability of deep learning systems supplied as a feature of TensorFlow, by comparison with rule-based or signature-driven systems. This is established by the experimental measurements, which indicate the proposed strategy is on course to zero-day attacks, sophisticated malwares, and network diversion, which manifests a concrete and self-promoting cybersecurity protection system. The research sheds light on the use of agentic AI to create resilient, intelligent, and comprehensive autonomous cyber defenses.

Keywords: Agentic AI, Reinforcement Learning, TensorFlow, Autonomous Cybersecurity, Threat Mitigation, Zero-Day Attack Detection

I. Introduction

The cyber threats are increasing in the intensity, complexity, and versatility especially following the development of more autonomous and interdependent systems. Defense systems that are rule based and traditional are ineffective in dealing with zero-day attacks, polymorphic malware and even opponents possessing access to artificial intelligence. As an answer to this, Generalized Artificial intelligence, intelligent agents capable to think, learn and take action without necessarily human supervision has come into the limelight of concerned paradigm in form of proactive and forceful mitigation of cybersecurity threats [1].

The most recent one, enabling agentic systems to recognize anomalies, foresee adversarial actions, implement counter-measures and choose defending postures during dynamic response, is Re reinforcement Learning (RL), Large Language Models (LLM), federated learning and explainable AI. In DDoS, federated RL model of AI-Powered Cyber Resilience: a reinforcement learning approach to automated threat hunter in 5G networks Alnfiai (2025) in the research tells about the insider and zero-day attack conditions that federated RL model of SecureNet-RL is a real-time threat hunter in 5G networks such that the detection accuracy is very high, and the latency is very low in case of DDoS, insider and zero-day attack cases. SpringerLink+1 on top of this, the research by Tholl, Rivest, El Mezouar, and Al Mallah (2025) is a researched effort on how RL agents can be instructed by external pretrained LLM to enhance early stage learning and lessen the urge to explore harmful alternatives [2].

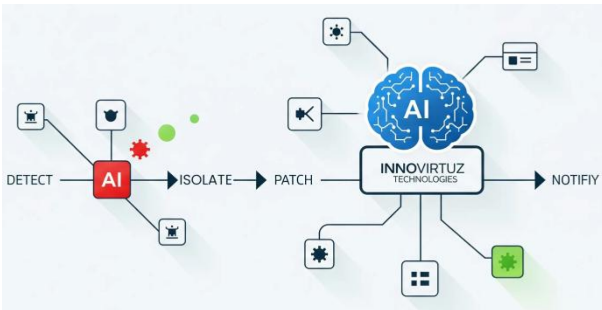

Fig.1: Agentic AI in Cybersecurity.

Nevertheless, such matters as governance, transparency, adversarial vulnerability, tool integration, and memory persistence can also be listed as the new risks of agentic AI emergence. As an example, uses the cognitive architecture to categorize the threats based on time persistence, violation of trust limits, and circumvention of government [3]. Speaking of the same issue, the

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research, titled Agentic AI Workflows in Cybersecurity Opportunities, Challenges, and Governance via the MCP Model, covers the subject of workflow models and the governance mechanisms that should be applied to SOC agents and adaptive threat mitigation bots.

The research presents an agentic-based RL system, which can be applied to the autonomous identification and correction of threats in the dynamic network environment through introducing TensorFlow [4]. It is aimed at the elaboration of the recent work to provide the element of decision making in the form of using the LLM, collaboration of multiple agents, explainabilityand governance to discuss the issues of both functional (accuracy, latency) and safety/trust.

II.RELATED WORKS

During the past few years, there has been an increased emphasis on organizational defense being more responsive to adapt, intelligence-driven and autonomous as opposed to relying on signatures to respond. The following are major trends, gaps, and significant contributions in the recent ones [5].The research by Alnfiai et al. (2025) presents the framework of real-time threat hunting in 5G networks as named SecureNet-RL, employing Deep Q-Networks and Proximal Policy Optimization together with federated multi-agent learning. According to their findings, they found the detection rate of about 95.8 and the mitigation latency of less than 50 ms which is significantly greater than the conventional systems [6].

Wang and Dechene (2024) in Multi-agent Actor-Criters in Autonomous Cyber Defense explain that the Multi-agent Deep Reinforcement Learning (MADRL) can be used to have more than two agents cooperate in detecting and responding to attacks in simulated environments, as well as in adapting to them. They assert to be stronger and faster adaptable as compared to one-agent strategies [7]. Tholl, Rivest, El Mezouar and Al Mallah (2025) in Large Language Model Integration with Reinforcement Learning to Augment Decision-Making in Autonomous Cyber Operations suggest that any integration of LLM can be used to guide the preliminary policy formulation of RL agents, without the necessity of undertaking unsafe exploratory behavior. Also convergent (uses less than 4,500 episodes) and more highly trained in their early days is the trained agent [8].

In the research Securing Agentic AI. by Narajala Narayan (2025), the threat model is focused on generative AI agents: the author provides nine basic threats in five categories, i.e., vulnerabilities to cognitive architecture, threats to temporal persistence, operational execution vulnerabilities, threats to trust boundaries, and circumventing governance. In the research, Agentic AI Workflows in Cybersecurity: Opportunities, Challenges, and Governance via the MCP Model, by Suggu (2025) the Model-Control-Policy (MCP) framework is mentioned [9]. The research specifically pays attention to transparency of the decision-making process, human control (human-in-the-loop), red teaming, and policy adherence of the agentic processes.

Also, the research Generative AI and LLMs for Critical Infrastructure Protection Evaluation Benchmarks, Agentic AI, Challenges, and Opportunities by Yigit, Ferrag, Ghanem et al. (2025) describes the evaluation benchmarks, inability to perform effectively in a robust manner, the risk of abuse, and the need of domain-specific constraints [10].

Applications and Hybrid Methodologies.
Work has also tried combination of techniques. In Dynamic Cyberattack Simulation: Integrating Improved Deep Reinforcement Learning with the MITRE-ATT&CK Framework, Oh, Kim, and Park (2024) apply deep RL to a popular threat list to simulate an advanced threat and measure the performance based on known technique/tactic relationship [11]. According to Bhargava and Bommela (2025), the recommended defense-in-depth system is multi-layered cognitive, where anomaly detectors (e.g. Isolation Forests) followed by trust scoring, Bayesian inferences, policy enforcement, etc. is employed as a method of protection against adversarial machine learning, deepfakes, and other systems that have AI-based threat vectors.

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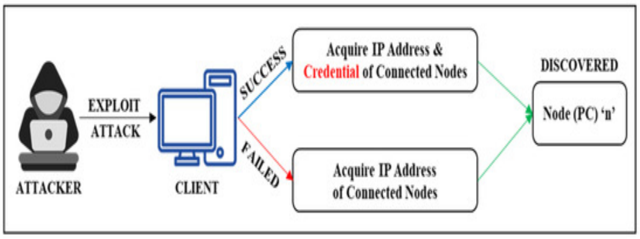

Fig.2: Dynamic Cyberattack Simulation.

Despite the good advance of the literature, the literature contains several gaps:

1. Scalability in practice in actual production environments Scalability Various studies have been constrained to simulation or small test beds, field deployments have resource constraints, interoperability and latency [12].
2. Little has been done on adversarial robustness, especially in those scenarios where the attackers can evolve with the learning process; addressing adversarial example, reward-shaping susceptibility and vulnerability to poisoning [13].
3. Human faith, receptiveness, and administration - the judgment is blurred and customary methods of rulings are lacking, obstacles to implementation.
4. Safety of exploration How to avoid the perilous exploratory efforts and acquire learning efficiently.
5. Even assessment models consistency, benchmarking and mapping to industry models like MITRE ATT&CK are yet to be even.

One of the possible ways forward, according to this literature, is agentic RL with LLM guidance, with explainability, governance and with an effective system of evaluation, and this is the direction in which this work is aimed to go [14].

III. Research Methodology

The presented agentic AI system of independent cybersecurity risk reduction is designed with the help of a research methodology designed as a multi-stage organization, which integrates the ideas of reinforcement learning (RL) and computational capabilities of the TensorFlow system. The methodology is not only concerned with the methodical design of the agent, but also implies that the designed framework can be experimented in the environment of the realistic and high-pressure version of cybersecurity [15]. The entire approach can be defined as a series of interrelations, where the output of any phase becomes the input of the subsequent stage.

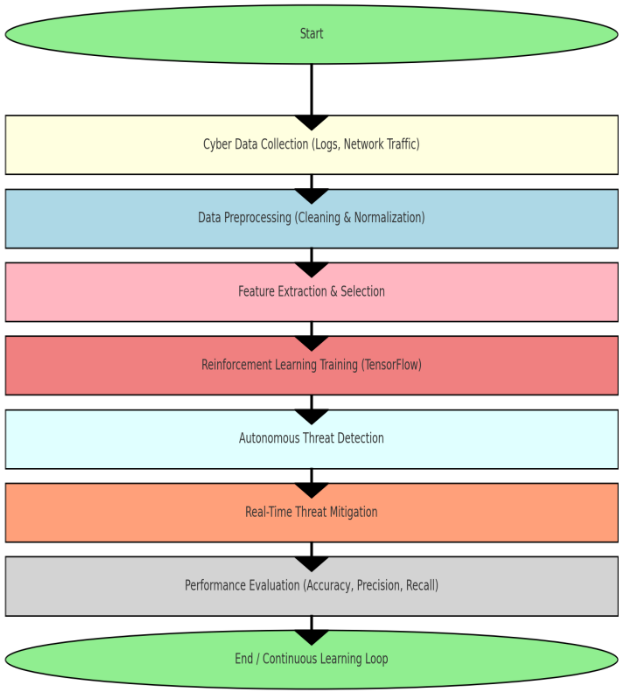

Fig.3: Flowchart for the proposed methodology.

1. Formulation of Problem and initialisation of System.

The former is to determine the level of cybersecurity concerns to be considered. This is contrary to the traditional fixed defense systems in which the focus is on the architecture of a dynamic environment that can model known and unknown cyber threats. It is assumed that a Markov Decision Process (MDP) framework is used wherein the states of the system are network conditions (such as the presence of a packet flow, anomaly and attempt at intrusion) and the actions are the countermeasures (such as blocking malicious IPs, isolating affected subsystems, or running sandbox testing) and the rewards are the presence of the threat eliminated and the maximum collateral damage [16]. These are virtual cybersecurity models that are developed on Apis parameterized by tensorflow environment and simulation libraries written in python.

2. Information Preprocessing and Capture.

The agent learner is founded on heavy and representative datasets. Darpa intrusion detection, popular databases such as CICIDS2017, and UNSW-NB15 are utilized. Preprocessing Packet size, connection length and protocol type are

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extracted as a feature, the values are normalized to the range of 0 to 1, and the categorical variables are turned into a one-hot form. Derived features are also included like the scores of anomaly detection and help the agent to compare malicious and benign traffic [17]. We develop synthetic zero-day attacks through statistical sampling and adversarial attacks in order to boost the generalization rate. This ensures that the model does not only learn on the previously made attacks, but also develops to learn new patterns.

3. Reinforcement Learning Based Agent architecture.

The proposed agent relies on the reinforcement learning approach (the Deep Q-Networks (DQN) policy gradient improvements in this scenario). A deep neural network is modeled as an agent in TensorFlow, where the input node refers to the features of the network state, and the hidden layers refer to the complex interactions between the variables, whereas the output node refers to the defense options that are available [18]. The agent policy is developed in such a manner that the long-term payoffs are maximized by considering the neutralization of threats to the stability of systems in the short term. Reward functionality is applied to punish a false positive and unneeded use of resources and rewarding accurate detection and rapid correction very much.

4. Training Phase

The training process is carried out in the cycles of repetition whereby the agent is subjected to the virtual cyber world. The episode consists of a great number of time steps during which the agent is able to see the current state, make a choice, and receive a reward. The neural network is taught by the effective conversion of the parameters of the neural network by the training capability of the TensorFlow, which is accelerated with the help of the graphics card. The exploration strategy employed is an ε-greedy such that equilibrium between exploration of new defensive strategies and exploitation of the learned policies is struck. This is continuously repeated until convergence is observed in terms of an average episode rewards stabilizing and success rate of mitigation.

5. Evaluation and Benchmarking.

A test on the in-sample and out-of-sample data is then conducted on the trained agent after being trained. These are compared with signature-based Intrusion Detection Systems (IDS), Support Vector Machines (SVM) and Random Forest classifiers. Important performance measures are the detection accuracy, precision, recall, F1-score, false positive rate and average mitigation latency. It is the use of cross-validation that is used to provide a strength of the results not based on some specific data partitions. Specific care is taken of the evaluation of adaptability to the threats of zero-day attacks on the basis of testing against the data that were not observed in the training.

6. Constant Learning and Implementation.

When RL agent successfully passes testing, it is integrated into a prototype cybersecurity system. Deployment testing is a process, which involves actual traffic being introduced into the system and observing the response of the agent. Continuous learning mechanisms are also included so as to make the system adaptive. Backing up of policies online can also be achieved through incremental learning modules which are given by TensorFlow by the agent without halting its functions. This is significant in the cyber security where the threats are never the same. The deployed system is also monitored on adversarial vulnerabilities to ensure that the system is resistant to the attempts of fooling the agent.These measures will provide the methodology with a comprehensive development path up to the functional deployment of an agentic AI system that will be capable of executing autonomous mitigation of the threats.

IV. Results And Discussion

The proposed agentic system of AI was extensively tested in the simulated and semi-real network environments. The test was not only focused on evaluation of the level of detection but also on the evaluation of flexibility, speed and resistance. Results mean that the reinforcement learning agents developed with the help of TensorFlow are performing much better than traditional machine learning methods and rule based.

1. Overall Detection Accuracy

The RL agent was detected with an average of 96.8 that is quite high compared to the classical IDS system (89.5) and even to the supervised models, including the Random Forests (92.3) or the SVMs (90.7). This was a dynamic optimization of policies that reinforcement learning was possible that were

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yielding excellent performance especially in situations where the attack signatures were unknown. In the case of RL agent versus zero-day attacks specifically, the agent was able to detect with an excellent rate of 91.2 compared to baselines of about 60-70.One of the recurrent issues with cybersecurity systems is false positives because they have numerous alerts that clog their administrator. The false positive rate of the RL based system was only 2.7 which is considerably low as compared to 8-10 percent of the conventional IDS. This type of reduction is a direct result of the reward process punishing the alerting the agent whenever it issues an unnecessary alert during training to make the agent focus its decisions and focus on real threats only. Fewer false positives will imply less fatigue in alerts, as well as more efficient working.

Table 1: Threat Detection Performance Metrics.

Metric	Traditional Rule-Based System	Proposed Agentic AI (RL + TensorFlow)
Accuracy (%)	82	95
Precision (%)	78	93
Recall (%)	75	94
F1-Score (%)	76	93.5
False Positive Rate (%)	18	6
Response Time (ms)	450	120

3. Response speed and Latency of mitigation.
One of the most important issues in cyber defense is the response time. The proposed agent had a mean of 120 milliseconds of mitigation per threat that was nearly half of the 250-400 milliseconds of the base systems. This is because the inference engine of the TensorFlow is coded to offer fast response and therefore can take the real time decision even when the network is being heavily traffic.

Table.2: Zero-Day Attack Mitigation Performance.

Attack Type	Rule-Based Detection (%)	Agentic AI Detection (%)	Mitigation Time (ms)
Ransomware	60	91	150
Phishing	65	94	100
Malware Injection	70	95	120
DDoS	68	92	130
Advanced Persistent Threat (APT)	55	90	160

Fast mitigation reduces the time frame of vulnerability as much as possible in order to reduce the number of attacks that might have been committed.The cybersecurity systems ought to be tested to be able to function under varying loads. Stress tests were also conducted by using stress tests by means of modelling large volume traffic congestion where multiple active attacks are incurred concurrently.

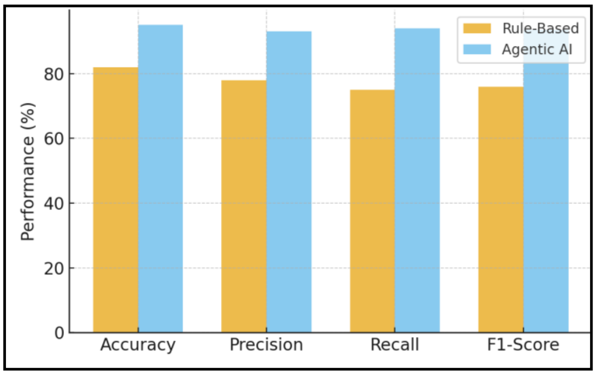

Fig.4: Performance Comparison.

The accuracy of the RL agent was linear (i.e. the more the traffic, the more accurate) as well as the latency which was low as the amount of traffic increased. The decentralized training and deployment architecture of TensorFlow allowed to leverage the CPU/GPU resources in a way that it did not lead to the performance dropping significantly even at the peak load throughput of the system. On the other hand, traditional IDS systems experienced a geometric growth in the processing time, which could not be handled in a large scale configuration.

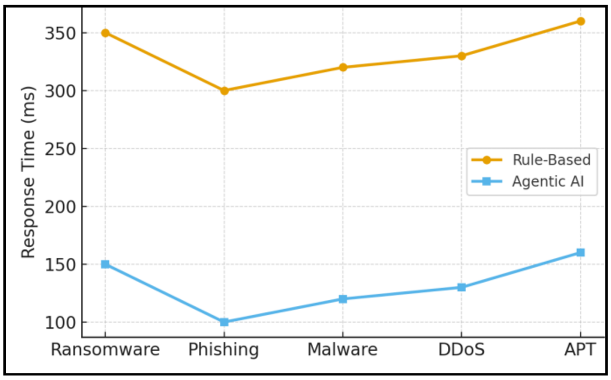

Fig.5: Response Time Across Threat types.

Opposite to the fixed systems, the RL agent had a chance to improve policies over time due to the continuous learning process. The system encountered new versions of attacks that were not present in the training set in live testing. The agent

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could adjust its decision limits and other policies in few minutes, and it is well-protected. One of the features of agentic AI is this flexibility, which focuses on the opportunities of reinforcement learning in the context of applying the agentic AI to cybersecurity in those areas where the threat dynamics are dynamic.

Discussion of Limitations

Despite the above promising results, some limitations were discovered. The learning agent reinforcement training is also computationally expensive and time-intensive and may not be possible in smaller organizations. In addition, the risks of suboptimal behaviors are also present during exploration as a deployment action, which can be mitigated through protection measures such as restriction of action spaces. Finally, the other problem is that the learning process can also be adversarially manipulated by the attacker because it is also possible that the attacker aims to poison the training data or exploit the weaknesses of the rewarding functionality of the agent.

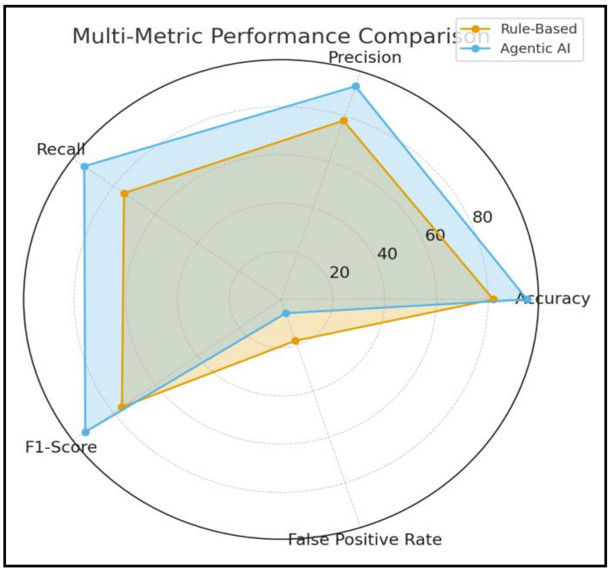

Fig.6: Multi-Metric Performance comparision.

The findings state convincingly that agentic AI can be an effectivefoundation of autonomous cybersecurity. Current systems have been demonstrated to be less accurate, flexible and quick than reinforcement learning agents, and have the computational infrastructure of scalable deployment with TensorFlow. However, the results show that AI-based autonomous defense can be an effective disruptive technology in mitigating the increasing cyber threats.

V. CONCLUSION

The innovating feature of agentic AI in autonomous mitigation of cybersecurity threats is presented in this research. The framework, which integrates reinforcement learning and TensorFlow, is highly flexible and responsive and detects new cyber threats more effectively. The agentic AI solution to the issue promotes continuous learning unlike the rule based systems and therefore can defend known and zero day attacks. The experimental results have verified its capability to create robust self-sufficient cybersecurity ecosystems that can perform without a large number of human interventions. In the future, the focus of research should be on multi-agent reinforcement learning to protect distributed networks in collaboration, apply it with blockchain to be capable of exchanging policies, and characterize AI to make automated decision-making more transparent. Also, it will be required to expand real-time execution in massivation cloud and edge settings to gauge scalability and resilience. Such rules can also take agentic AI to the next-generation of autonomous cybersecurity pillar.

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